Sahil Janjua

Software Engineer | Java, Spring & Cloud

Surrey, BC • sahil-janjua@hotmail.com

Profile Summary

• **Java Software Engineer** with **3 years** of experience building distributed backend systems, **REST APIs**, secure authentication systems, **AI**-powered resume analysis tools, and forms management platforms at **SAP** and **The Growth Spark**.

• Experienced across the **Java**/**Spring** ecosystem, including modern **Java** (**Java 21**, **Project Loom** virtual threads), **Spring Boot**, **Spring Security**, **Spring Cloud Gateway**, reactive programming, **Resilience4j** resilience patterns, **Micrometer** instrumentation, and **Maven**-based builds.

• Experienced in cloud-native and distributed systems development using **AWS**, **Docker**, **Kubernetes**, **Terraform**, **Apache Kafka**, **MongoDB**, **Redis**, and **SQL**, including **RESTful** backend services, containerized deployments, infrastructure as code, and production issue resolution.

Technical Skills

Programming Languages: Java, JavaScript/TypeScript, C++, Python, SQL, HTML/CSS
Back-End Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Spring Cloud, Spring Cloud Gateway, FastAPI, Node.js, Express.js
Front-End Frameworks: React, Next.js, Angular, Vite, Tailwind CSS
AI/ML & LLMs: LLM APIs, Transformer Embeddings, Semantic Similarity Search, Prompt Engineering, Vector-Based Candidate Matching
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis
Messaging & Real-Time Systems: Apache Kafka, WebSockets, Server-Sent Events (SSE)
Testing Frameworks: JUnit 5, Mockito, AssertJ, MockMvc, Jest, Jasmine, Selenium, JMeter
DevOps & Infrastructure: Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI, Jenkins, Prometheus, Grafana
Cloud Platforms: AWS (EC2, S3, IAM, VPC, CloudWatch, Polly), GCP
Developer Tools: Git, GitHub, Maven, Gradle, OpenAPI/Swagger, Jira, Postman, Chrome DevTools

Work Experience

The Growth Spark

Vancouver, BC

Junior Software Engineer

May 2024 - Present

• Built secure authentication services in Java and Spring Boot for AI resume analysis and forms management platforms, reducing support tickets and accelerating candidate screening.

• Decreased support tickets by **30**% through **Spring Boot** and **Spring Security** authentication workflows using **JWT** generation, secure cookies, **BCrypt** hashing, refresh token rotation, **Google OAuth 2.0**, and email verification.

• Created a forms management platform with **REST APIs** for validated submissions, **CSRF** protection, rate limiting, authenticated endpoints, file uploads, standardized error responses, role-based approvals, and **Google Drive** integration.

• Designed the **GPU** inference backend on **EC2 G4dn Spot** capacity using **Auto Scaling** groups, pre-baked **AMIs**, and interruption handling, maintaining availability while lowering inference costs compared with on-demand capacity.

• Reduced manual deployment effort by **60**% by containerizing backend **APIs** and **AI** inference services with **Docker** and **Kubernetes**, using **GitHub Actions**, **GitLab CI**, multi-stage builds, rolling updates, promotion gates, and test automation.

• Established logging, monitoring, and alerting with **Logback**, **Log4j2**, **CloudWatch**, **Prometheus**, **Grafana**, and **Slack** and email notifications to track **API** errors, **GPU** usage, and authentication failures while resolving production incidents.

• Raised code coverage to **78**%, caught **15+** regression issues, and reduced manual QA effort by **35**% by building test suites with **JUnit 5**, **Mockito**, **AssertJ**, and **MockMvc**.

SAP

Vancouver, BC

Software Developer Intern for SAP Analytics Cloud

Jan. 2022 - Sep. 2022

• Developed the **SAP Analytics Cloud** data blending widget, supporting an enterprise **BI SaaS** serving **30,000+** customers across **150+** regions and processing billions of rows daily in multi-tenant production environments.

• Resolved **5+** monthly production issues and increased client satisfaction by **15**% by debugging a **Maven**-based data blending widget using **Java**, **JavaScript**, **CSS**, **Node.js**, **Chrome DevTools** breakpoint/network profiling, multithreading/concurrency debugging, and **JVM**/heap profiling.

• Reduced render time from 8 seconds to 3 seconds on datasets exceeding 1 million rows for 1,000+ concurrent enterprise users by implementing virtualized table rendering, batching queries, and optimizing Java stream processing.

• Wrote and maintained unit, integration, end-to-end, **UI** automation, and regression tests with **Jasmine**, **Selenium**, and **JUnit**, reaching **82**% test coverage.

• Cut cycle time by **2** days by reviewing sprint code to catch bugs and receive feedback, running sprint demos for cross-functional stakeholders, and improving development practices.

Projects

Neuragate

• Implemented confidence-gated self-healing through a **Java 21 API** gateway built with **Spring Cloud Gateway**, **Project Loom** virtual threads, a **Kafka** telemetry/load-testing pipeline, real-time anomaly detection, an **AI** advisor, **Resilience4j** circuit breakers, and **Redis**-backed rate limits.

• Built reactive **JWT RBAC** for **3** roles, **11 Micrometer Prometheus** gauges, an **SSE** dashboard, and a stress harness generating **1,000** requests per minute using **Docker Compose**, **Kafka**, **Redis**, **Prometheus**, and **Grafana**.

• Engineered a configurable **fault-injection service** with tunable per-endpoint failure rates and injected latency, exercising **circuit-breaker** trip, fallback, and recovery paths under controlled chaos to validate the self-healing loop end to end.

• Exposed a custom **Actuator HealthIndicator** reporting gateway and downstream-dependency status, providing the liveness/readiness signal for container orchestration and **load-balancer routing**.

EcoBrief

• Kept **AWS** spend near zero by building a scale-to-zero distributed **AI** news oracle with **Java RSS** ingestion to an **S3** data lake, a **FastAPI** orchestrator, and **Boto3**-provisioned ephemeral **ARM EC2** instances that self-terminated after inference.

• Fit a quantized **3**-billion-parameter **Llama 3.2** model into **2 GB RAM** using **Ray Serve** and llama-cpp-python, generating **MP3** summaries with **AWS** Polly and serving transcripts and audio briefings through **React**, **Vite**, and **Tailwind**.

Multimedia Review Platform (Letterboxd Clone)

• Enabled dynamic movie retrieval and multi-tag genre filtering by integrating the **TMDb API** with a **Node.js** backend, **REST** endpoints, and **Axios**, mapping query parameters to genre **ID**s with **AND**/**OR** logic.

• Improved user management **CRUD** and secure access by implementing **JWT** authentication with **HS256** signing, token expiration, refresh token rotation, **Mongoose** validators, and middleware-driven request handling.

• Accelerated and stabilized **MongoDB** queries by designing an embedded, denormalized schema with compound indexes on userId, reviewId, and genre, plus text indexes supporting search functionality.

• Delivered real-time platform updates by configuring **Socket.IO WebSocket** rooms to broadcast user join and leave events and live review activity between clients and the server.

Education

British Columbia Institute of Technology (BCIT)

Sep. 2026 - Expected 2028

Master of Science in Applied Computing | Academic Focus: Artificial Intelligence & Machine Learning

Burnaby, BC

Simon Fraser University

Sep. 2020 - Jun. 2024

B.Sc. in Software Systems

Vancouver, BC